A proof of the conjectured identity for OEIS A309878

Adrian Perez Fontelles
Independent researcher

26 August 2026

Abstract

OEIS A309878 carries a conjectured identity expressing its terms through the terms of A009545. We prove it. Each entry involved posts a generating function, so the claim is a single identity between generating functions: writing F_a for the generating function of A309878 and F_j for those of the entries on the right, the conjecture says that a specific combination of the F_j, shifted and weighted as the identity prescribes, differs from F_a by a polynomial. That difference is computed here in closed form and equals 0, of degree -1, which proves the identity for every $n > -1$.

2020 Mathematics Subject Classification. 11B83, 05A15.

1 The sequences and the conjecture

OEIS A309878 is "The real part of b(n) where b(n) = (n + b(n-1)) * (1 + i) with b(-1)=0; i = sqrt(-1)". It has offset 0 and begins

$$a(0), \ldots, a(7) \;=\; 0, 1, 2, 1, -4, -13, -22, -23, \;\ldots$$

Its generating function, as posted on the entry, is

$$F_a(x) \;=\; \sum_{n\geq 0} a(n)\, x^n \;=\; -\frac{x\,(2x-1)}{(x-1)^2\,(2x^2-2x+1)}, \tag{1}$$

and the entry carries the following comment:

> **Conjecture:** a(n) = 2*A009545(n)-n. - *R. J. Mathar*, Mar 06 2022

As of the "Last modified" line on the live entry (2024-10-14, revision 37) the statement is still recorded as a conjecture, and no proof appears among the entry's links, formulas or programs.

The entries appearing on the right-hand side post the generating functions

$$F_{A009545}(x) \;=\; \frac{x}{2x^2-2x+1}$$

2 Shifts of a generating function

Throughout, a sequence b defined on the indices its OEIS entry uses is identified with the formal series $F_b(x) = \sum_{m\geq 0} b(m)x^m$, where $b(m) = 0$ for m below the entry's offset. This convention makes shifting immediate.

Lemma 1. *For an integer $k \geq 0$,*

$$\sum_{m\geq 0} b(m+k)\, x^m = \frac{F_b(x) - \sum_{m<k} b(m)x^m}{x^k}, \qquad \sum_{m\geq 0} b(m-k)\, x^m = x^k F_b(x).$$

Proof. Both are re-indexings of the defining sum. In the first, subtracting the terms below x^k removes exactly the coefficients that have no counterpart after the shift, and dividing by x^k relabels $m+k$ as m. In the second, multiplying by x^k relabels m as $m+k$, and the coefficients of x^m for $m < k$ are $b(m-k) = 0$. $\square$

Lemma 2. *Let $\theta = x\,\dfrac{d}{dx}$, so that θ acts on x^m as multiplication by m. For a polynomial c, the sequence $m \mapsto c(m)b(m)$ has generating function $c(\theta)F_b$. In particular $m \mapsto m^j$ has generating function $\theta^j\left[(1-x)^{-1}\right]$.*

Proof. θ multiplies the coefficient of x^m by m, so $c(\theta)$ multiplies it by $c(m)$. The last statement is the case $b \equiv 1$. $\square$

Corollary 3. *Let $R(x)$ be the generating function assembled from the right-hand side of the conjecture by Lemmas 1 and 2. Then the conjectured identity holds for every $n > d$ if and only if $F_a - R$ is a polynomial of degree at most d.*

Proof. The coefficient of x^n in $F_a - R$ is the difference between the two sides of the identity at n. That difference vanishes for all $n > d$ exactly when $F_a - R$ has no terms above x^d. $\square$

Note that the test is for a *polynomial*, not for zero. A claim of this kind typically fails at a few small indices, where the shifted terms on the right reach outside the range of their own sequences; the polynomial records precisely those indices.

3 The computation for A309878

Assembling the right-hand side by Lemmas 1 and 2 gives

$$R(x) \;=\; -\frac{x\,(2x-1)}{(x^2-2x+1)\,(2x^2-2x+1)},$$

and subtracting (1) the rational parts cancel, leaving

$$F_a(x) - R(x) \;=\; 0,$$

a polynomial of degree -1.

Theorem 4. *The conjectured identity holds for every $n > -1$.*

Proof. Immediate from Corollary 3 and the displayed difference. $\square$

4 Verification

Every number above is an output of a script written separately from this note, and two independent checks were run.

First, no generating function was assumed. For each entry involved, the posted expression was expanded as a Taylor series and compared term by term with that entry's own DATA section; only expressions reproducing the published terms were used.

Second, the identity itself was evaluated directly on the published terms, in exact integer arithmetic and without reference to any generating function: for each n in range both sides were formed from the entries' own values and found to agree, for all 45 values of n from $n = 0$ upward. This is what Corollary 3 predicts from the degree of the difference, and it is a genuinely different computation from the symbolic one.

The symbolic step is exact throughout: the difference is obtained by rational-function cancellation, not by series truncation, so the displayed identity is an identity of functions and the theorem holds for all $n > -1$ simultaneously.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A309878.

[2] H. S. Wilf, *generatingfunctionology*, 3rd ed., A K Peters, 2006, Chapter 2 (series manipulations and shifts).

[3] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011, Chapter 1.